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# Intelligence Engineering

## Solving Real World Problems At Scale

Prof. Dr.-Ing. Mark Schutera | September 25, 2026 · *drifting stream*

LECTURE VIII

# Data Distribution Shifts and Monitoring

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# This lecture in one slide

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## VIII Data Distribution Shifts and Monitoring

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Causes of ML System Failures

Data Distribution Shifts

Monitoring and Observability

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*why this lecture matters, and what  
you can do after it*

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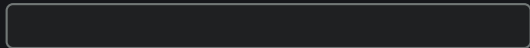
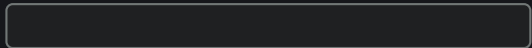
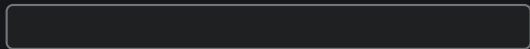
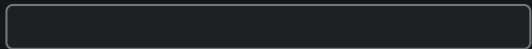
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# Causes of ML System Failures

# Software System Failures

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$P(y|x)$



[1]

# ML-Specific Failures

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$$P(\hat{y} | x)$$

$$[1]$$

# Production data differing from training data

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| Training data | Production data |
|---------------|-----------------|
|               |                 |

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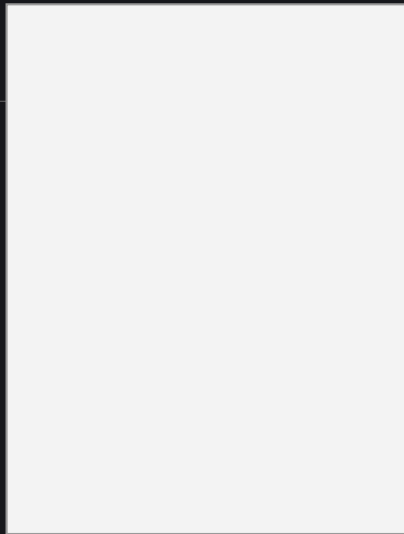
# Edge cases

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**Definition: Edge cases**

# Degenerate feedback loops

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$P(y | x)$

[1]

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# Data Distribution Shifts

# Types of Data Distribution Shifts

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$P(y | x)$

**Covariate shift**

**Label shift**

**Concept drift**

[1]

# Covariate shift

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**Definition: Covariate shift**

$P(y | x)$

[1]

# Label shift

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$$P(y|x)$$

$$[1]$$

# Concept drift

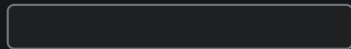
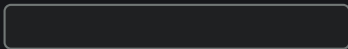
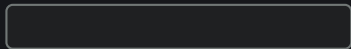
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**Definition: Concept drift**

# General Data Distribution Shifts

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$$P(y | x)$$



$$[1]$$

# Detecting Data Distribution Shifts

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$P(y | x)$

**Statistical methods**

**Time scale windows**

[1]

# Statistical methods

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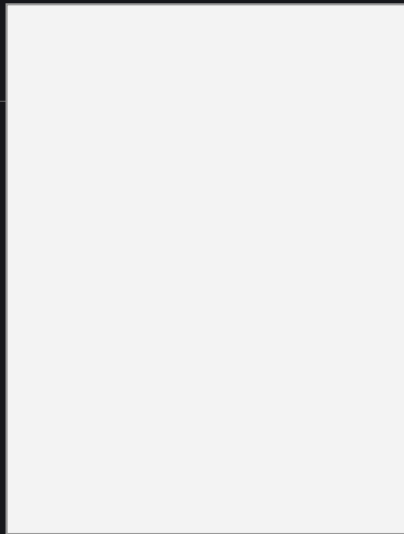
$P(y|x)$

Statistical methods

[1]

# Time scale windows for detecting shifts

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$P(y | x)$

[1]

# Addressing Data Distribution Shifts

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$$P(y | x)$$

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[1]

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# Monitoring and Observability

# ML-Specific Metrics

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$P(y|x)$

**Accuracy-related metrics**

**Features**

**Predictions**

**Raw inputs**

[1]

# Monitoring accuracy-related metrics

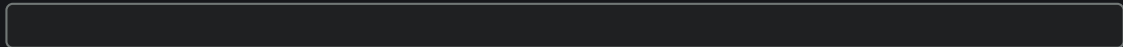
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$$P(\hat{y} | x)$$

$$[1]$$

# Monitoring predictions

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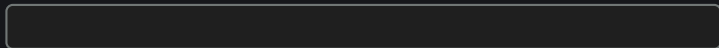
$P(\hat{y} | x)$

[1]

# Monitoring features

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$P(y|x)$



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# Monitoring raw inputs

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$P(y|x)$

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# Monitoring Toolbox

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**Logs**

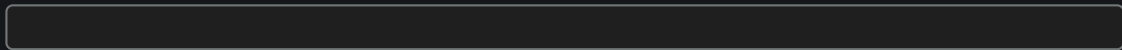
**Dashboards**

**Alerts**

# Logs

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$P(y|x)$



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# Dashboards

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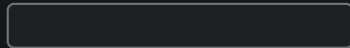
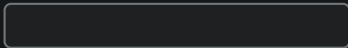
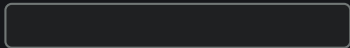
$P(\hat{y} | x)$

[1]

# Alerts

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$P(\hat{y} | x)$



[1]

# Observability

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$P(y|x)$

**Monitoring**

**Observability**

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# Summary

# Summary

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$P(y|x)$

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[1]

# References

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- 1 Chip Huyen. *Designing Machine Learning Systems: An Iterative Process for Production-Ready Applications*. Sebastopol, CA: O'Reilly Media, 2022.